Technical decisions and unresolved questions

Defect-Aware Multi-Die Image Fusion for Semiconductor Reference Reconstruction · Md Shamiul Islam Riyad · Section letters and [n] refer to the technical document.

Decisions taken (recommendations; each can be revisited)

Topic	Decision	Reason
Scenario	Same-design dies with complementary defects (S1) only	Repeated views of one die cannot remove its defects; wafer maps contain no surface images (B.1)
Output	Candidate reference $\hat{R}$ plus masks, support S, uncertainty U, abstention A, provenance	Users need to see where the reference is trustworthy; no claim of physical repair or electrical function
Strategy	Stage 1 classical → Stage 2 learned evidence → Stage 3 optional fill	A transparent baseline first; learned parts must beat it on held-out designs (F.4)
Registration	Phase correlation, then ECC (Euclidean/affine), defect-masked second pass	Sub-pixel, gain-invariant, low risk of absorbing defects; SIFT/ORB only with distinctive features (F.1)
Defect evidence	Cross-die MAD z-score baseline; U-Net, PatchCore or PaDiM added in Stage 2, calibrated	Pixel-level evidence is required; image-level classifiers give no masks (F.2)
Fusion	Weighted median over reliable dies; structure-aware single-source fallback	Robust and traceable; a plain median blurs edges, mixes structures, breaks down at ≥ 50% (G.4)
Missing evidence	Abstain (NaN + mask) when $S < s_{min}$ (initially 2) or SE is too large; learned fill off by default	Avoids hallucinated circuitry [44]; systematic defects are unrecoverable from images alone
Diffusion	Pretrained only, experiment only, outputs flagged; no training from scratch	Data far too small; generative fills optimise plausibility, not truth (F.3)
Data	Public data for evidence; synthetic benchmark for main metrics; real same-design groups for validity	No public dataset has same-design groups with a clean reference of the same die (D.1)
Splits	By design / layout family / acquisition group, with hashed manifests	Prevents near-duplicate leakage between train and test (D.6)
Primary metrics	Defect leakage DLR, silent pass-through SPR, edge F-score, PSNR on common pixels, with coverage	Fidelity alone rewards blur; abstention must not inflate accuracy (J.4)
Hardware	CPU for Stage 1; 8–12 GB GPU for training; 16–24 GB+ only for large or diffusion work	Estimates; confirm with a profiling run before buying or booking (H)

Unresolved questions

1. **Novelty.** Do published or patented methods already combine per-die reliability, weighted fusion and explicit abstention? Industrial multi-die median and computed references exist [15–17].
2. **Real data.** Can a lab or industry partner provide 4–5-die same-design SEM groups with verified clean or design-rendered references, under what licence or NDA?
3. **Registration on real SEM.** Is sub-pixel accuracy reachable under charging, drift and scan distortion? How often do periodic layouts cause one-pitch errors?
4. **Process variation.** Will normal die-to-die variation be flagged as defects, cutting support and coverage?
5. **Detector transfer (H5).** How far do U-Net, PatchCore and PaDiM trained on Carinthia-S, MVTec or synthetic data transfer to the target layer?
6. **Repeated defects.** Without design data, how can defects shared by all dies be detected at all?
7. **Thresholds.** Best s_{min}, τ_r, κ and leakage tolerance τ_ℓ; to be set on validation designs.
8. **Synthetic realism.** Which defect and imaging models make synthetic rankings predictive of real ones?

Facts still to confirm

1. Licences of Carinthia (not found), Carinthia-S (third-party: CC BY 4.0), VisA (conflicting), WM-811K and MixedWM38.
2. Carinthia record identity (concept vs. version DOI); Carinthia-S resolution and class semantics.
3. Missing DOIs or volumes: PaDiM, DRAEM, LaMa proceedings; Farsiu et al.; year of [10]; MixedWM38 paper.
4. Every link opened once by hand (the build environment could not reach publisher sites).

Cannot be validated without real data. Without verified clean references, PSNR, SSIM, leakage and edge metrics cannot be computed on real images. Without real same-design groups, every result stays synthetic. Without design data, defects shared by all dies cannot be told apart from intended structure. Electrical function is never validated.